Chase Golden – Assignment #2

Data Transformation and Simplification

For this project, the dataset was structured to represent TixScout's customers, which is a ticket price aggregation and predictive insights solution. TixScout helps customers find the best prices on tickets for concerts, sporting events, and live entertainment spaces, and the goal of the company's marketing team is to better understand the company's 5,000 customers so that they can better develop a target outreach campaign and increase engagement with their TixScout app. The primary business goal of TixScout is to enable the members of its platform to become more active users by converting customers from casual browsers to buyers who are committed to purchasing tickets. To do this, their marketing efforts will build connections based on segments that identify lifestyle-based categories that relate to actual event attendance and willingness to spend. Specifically, TixScout will be very interested in any sedentary habits customers have concerning digital engagement to compare their financial capacity or spending habits, and across lifestyle variables like pet ownership, any fitness engagement, and typical commute, where those variables will help TixScout determine customer participation in an event.

To better achieve the action objectives, the 60-column dataset was thoughtfully reduced to focus on variables that were prime for helping to understand ticket purchasing behavior, likelihood to attend events, and engagement with the event app. I retained features that were most directly related to demographics, lifestyle habits, technology adoption, and financial flexibility. Less useful, overly detailed, irrelevant to entertainment engagement, or duplicates were removed from the user data. Associated fields were grouped into their values for broader and more usable segmentation. For example, pet types were not retained, but rather just the total number of pets and a binary Pet Ownership to address pet-friendly events without adding complexity. Commute time values were grouped into a categorical Commute Category to emphasize the likelihood of attending weekday events. Annual income and debt-to-income ratio were adequately captured, but merged into one Financial Flexibility Score. Multiple technology and subscription service variables were merged into composite engagement measures, which included Tech Engagement Score, and Digital Service Package to assist with profiling purposes. Lifestyle indicators based on activity, streaming, and credit card usage levels were retained because of their meaningfulness for entertainment consumption levels.

The structure of the dataset allows for each transformation to correspond to TixScout's business objectives and ensures that a prepared dataset is ready for meaningful marketing analytics and customer segmentation processes in the next analysis stage.

1 Important Analysis and Removing Noise

The original dataset had a staggering 60 columns of customer characteristics, ranging from core demographics to incredibly detailed service usage details. The original data certainly had all the bases covered, but much of it was too specific, repetitive, or just irrelevant to TixScout's business objectives. During the process of cleaning the data, I tried to retain variables that could be useful for segmentation for marketing and predicting post-event attendance, while still losing the noise and merging together enough similar fields to arrive at the most usable dataset.

These comprised demographics and household-context factors, including region, town size, gender, age, education years, job type, time employed, retirement status, marital status, and household size. Financial capacity indicators, including annual household income, income-to-debt ratio, loan default indicator, credit card and other debt amounts, credit card types, credit card time, credit card monthly items, and credit card monthly spending, were retained for a better understanding of spending power and purchasing behavior. Lifestyle and time availability,

including commute time, physically active lifestyle, and pets, were retained to assess their role in participation in events. Examples of technology engagement measures were tenure with telecom provider, internet or data plans, device ownership, any subscription services, any streaming activity, and smart home use, to assess customers' digital readiness, and their likelihood to engage with TixScout's app.

Additionally, there were also some areas of the data that I decided to eliminate completely, as they were simply adding complexity to the segmentation without providing any fundamental segmentation value. For example, there were specific pet counts that I could conveniently replace with a total pet count. Other segments were eliminated. Political affiliation, voting history, and idiosyncratic voice usage or device details, were also eliminated, as there did not seem to be a pivotal relevance with the ticket purchasing behavior of the attendees.

To ultimately combine the data fields and reduce the dataset, I grouped related columns under larger, more actionable categories. Pet data was grouped into binary Pet Ownership, commute times were shaped into a Commute Category, and financial stress was grouped into Debt Stress Level. Financial readiness was captured through a composite Financial Flexibility Score that captured income and debt stress, while technology adoption was summarized in a Tech Engagement Score. I aggregated streaming behavior as Streaming Intensity, premium services and associated usage in a Premium Services Count, credit card activity into a Credit Card Usage Level, and groupings that summarize bundled tech products in a Digital Service Package.

The result of the purposeful noise reduction is that the resulting dataset is now a purposeful one that is easier to read by people and matches TixScout's way of thinking about the data for marketing and segmentation purposes. Each column serves a clear purpose, and future analysis can be thoughtful and purposeful, without the need to continually refer to the original data dictionary.

2 Feature Creation

This phase of the project involved the development of new, supporting features that would help the dataset enhance TixScout's marketing and segmentation strategy. In total, I created ten new variables that suffice variability, a defined purpose, transformation logic, and methodology to condense segmentation and prediction of customer behavior.

The first was Pet Ownership. A binary field vetted off the total number of pets, customers who had greater than zero pets were flagged as "Has Pets" or "No Pets". The number of different pet types and use of multiple columns was simplified to a single variable, easy to filter allowing TixScout to target pet-friendly events without complication. Commute Category field is a new variable that transformed raw commute times into three bins. Short (0-15 minutes), Medium (16-45 minutes), Long (46+) to help highlight geographic convenience and likelihood of returning to weekday events.

Financial segmentation was also addressed with two variables. Debt Stress Level was able to group debt-to-income ratios into Low (< 20%), Moderate (20% - 35%) and High (> 35%) stress categories allowing for a targeted affordable strategy without exposing sensitive data. A snapshot of Financial Flexibility was achieved through a variety of variables designed to create a single indicatively of aggregate loop of affordability metrics. Score a customer's combined annual household income and debt-to-income ratio into a single number, which is the most complete way of supplying a person's potential earning power and financial obligations, which is key to recognizing customers who are both capable and motivated to buy premium-priced tickets.

Technology engagement and adoption was summarized with a Tech Engagement Score, which highlighted monthly data usage, smart devices, smart home devices, and dedicated home computer usage on a 0–10 scale. Level of engagement noted that digitally ready customers are more than likely to interact with TixScout App when launched. A Streaming Intensity measurement classified weekly streaming time into Light (<5

hours), Moderate (5–15 hours), and Heavy (>15 hours), and provided by a proxy to customers' entertainment interests and directions to make better recommendations of events.

Lifestyle targeting was enhanced with the Active Fitness Flag, which shows where active lifestyle customers had a 1 and other customers had a 0, which helps targeting with athletic or outdoor events for relevant audience segments. Premium Subscription Count addressed spending capacity and willingness, and represented how many premium subscriptions a user had, such as video streaming add-ons, cloud storage, smart home monitoring, and financial news streaming, counted to create a score between 0 and 4.

Credit Card Usage Level was another indicator of purchasing behavior, where cardholders were classified as Low, Medium, or High usage based on a clear designation to help frame purchasing frequency and value. Additionally, Digital Service Package Flag identified customers with bundled tech services or device-based services, including cloud backup, mobile security, or virtual assistants, who were flagged as "Has Package" or "No Package." This was useful in identifying tech-savvy customers who may be more responsive to integrated event notification functionality or multi-service ticket packages.

Taken as ten items reduced overlap, increased interpretability, and strict alignment with the core business goals of TixScout. Each was constructed for quick filtering and simple segmentation, established a clear connection with ticket-purchasing behavior, and ensured that TixScout had everything they needed for specific marketing analysis in the final project.

3 Qualitative and Graphical Analysis

The analysis began with five visualizations with one variable to assess the spread of major customer characteristics. The Pet Ownership bar chart illustrated that nearly all customers utilized at least one pet service. This finding demonstrates a significant opportunity to promote pending events catered to pet owners and to potentially incentivize partnerships with pet-focused service providers. The Commute Category bar chart illustrated that most customers have medium (16–45 minute), commute times, considering weekday events may better fit customers' travel time frames. The Debt Stress Level bar chart intuitively depicted that most customers, overall, are at a moderate level of financial stress: Enough to keep their engagement reasonably affordable, overall, but not zero. While assessing entertainment engagement, the Streaming Intensity bar chart illustrated that there is a significant proportion of customers who are considered heavy streamers, and this group is a great fit for promoting live entertainment and events. The Premium Subscription Count histogram illustrated that while many customers subscribe to at least one premium digital service, there is a slightly smaller number of customers who subscribe to all four services; this second group represents a valuable opportunity for developing VIP or exclusive ticket packages.

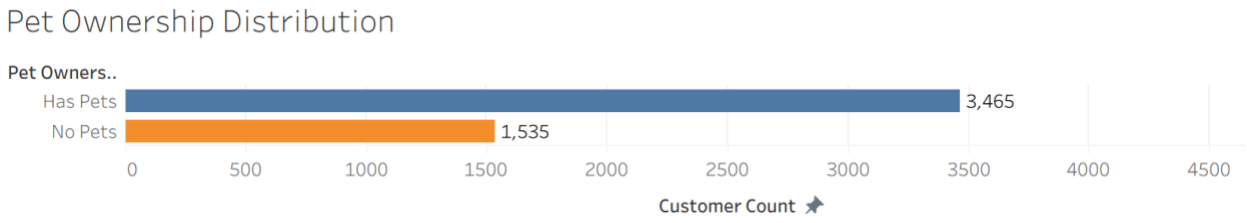


Figure 1: Pet Ownership Distribution

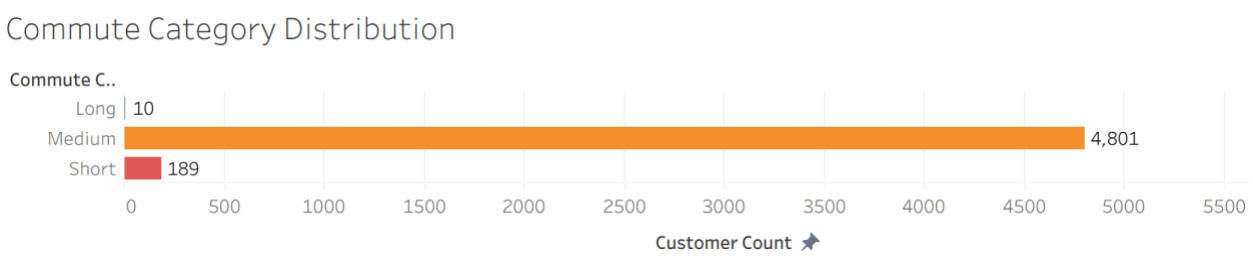


Figure 2: Commute Category Distribution

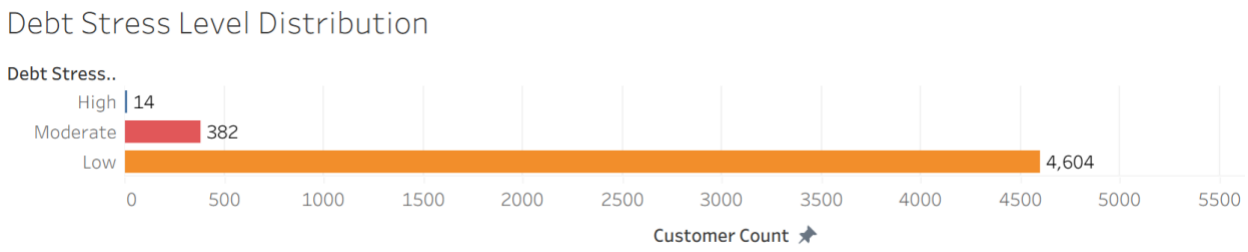


Figure 3: Debt Stress Level Distribution

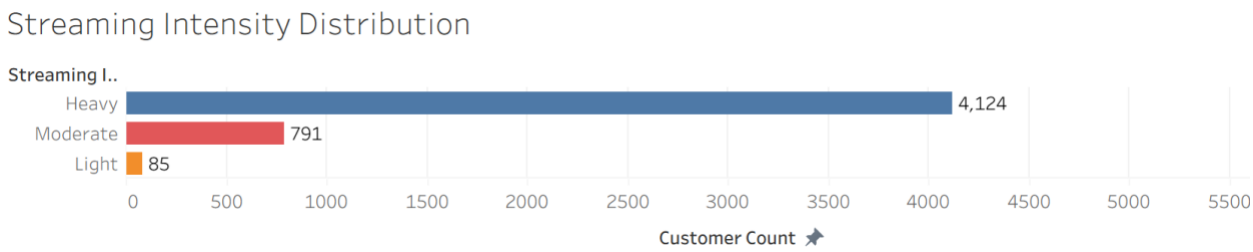


Figure 4: Streaming Intensity Distribution

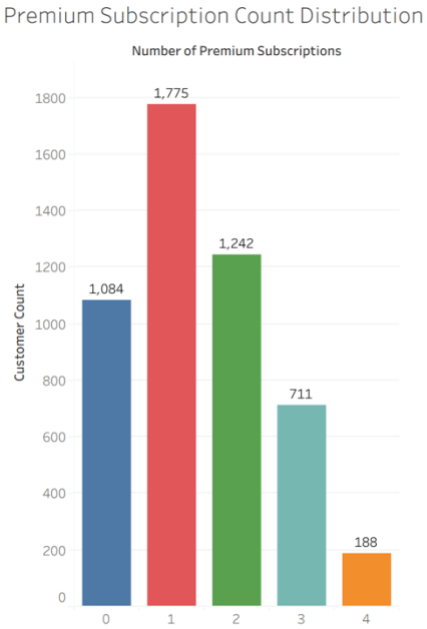


Figure 5: Premium Subscription Count Distribution

The visualizations with two-variable plots revealed valuable details concerning the overlap in customer characteristics. Specifically, with the Tech Engagement Score vs Streaming Intensity grouped bar chart, it was easy to see there is a strong link between heavily streaming users and having a higher tech engagement score, which can highlight a digital marketing opportunity. In the Debt Stress Level vs Financial Flexibility Score scatter plot, it was evident that lower debt-stressed customers tend to have higher financial flexibility and may yield a great target group for high-priced ticket offers. The Premium Subscription Count vs Credit Card Usage Level stacked bar chart made a clear association between higher credit card usage and more premium subscriptions, which is an active spending group that likely will yield additional upselling opportunities. The Commute Category vs Active Fitness Flag clearly illustrated a higher presence of medium-commute customers in both active and inactive groups, which indicates that this distance is the sweet spot for most marketing campaigns. The pet ownership vs streaming intensity chart also demonstrated that heavy streamers, particularly, were more likely to have pets and, at the same time, presented some possibilities for combined lifestyle and entertainment-themed marketing efforts.

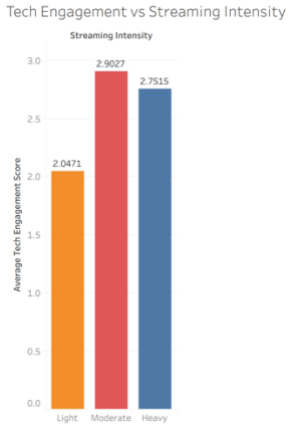


Figure 6: Tech Engagement vs Streaming Intensity

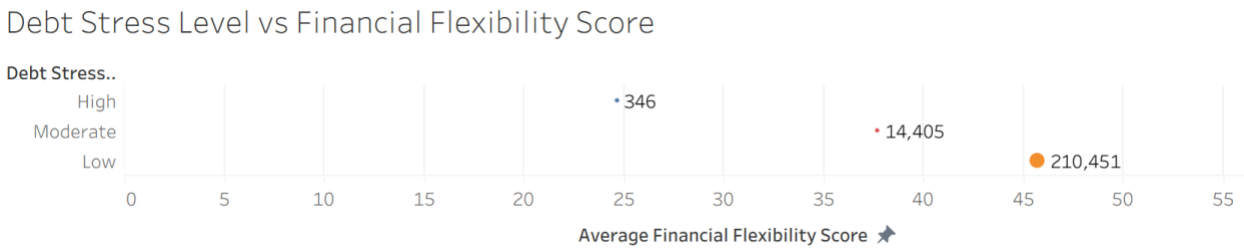
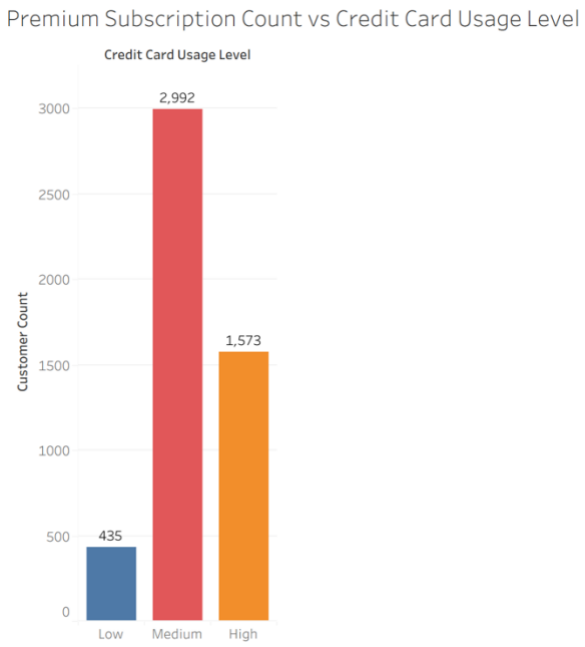


Figure 7: Debt Stress Level vs Financial Flexibility Score



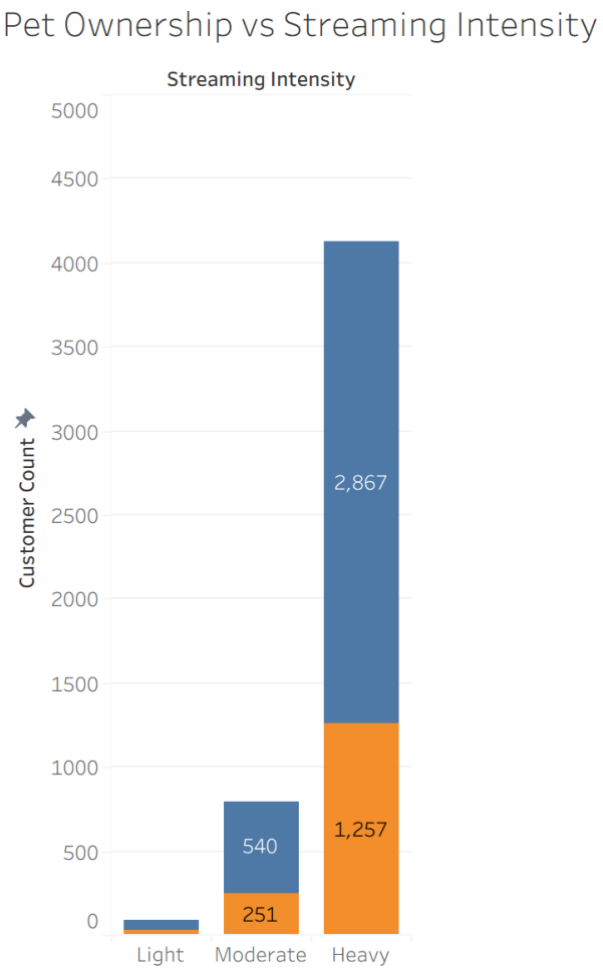


Figure 8: Premium Subscription Count vs Credit Card Usage Level

Figure 9: Pet Ownership vs Streaming Intensity

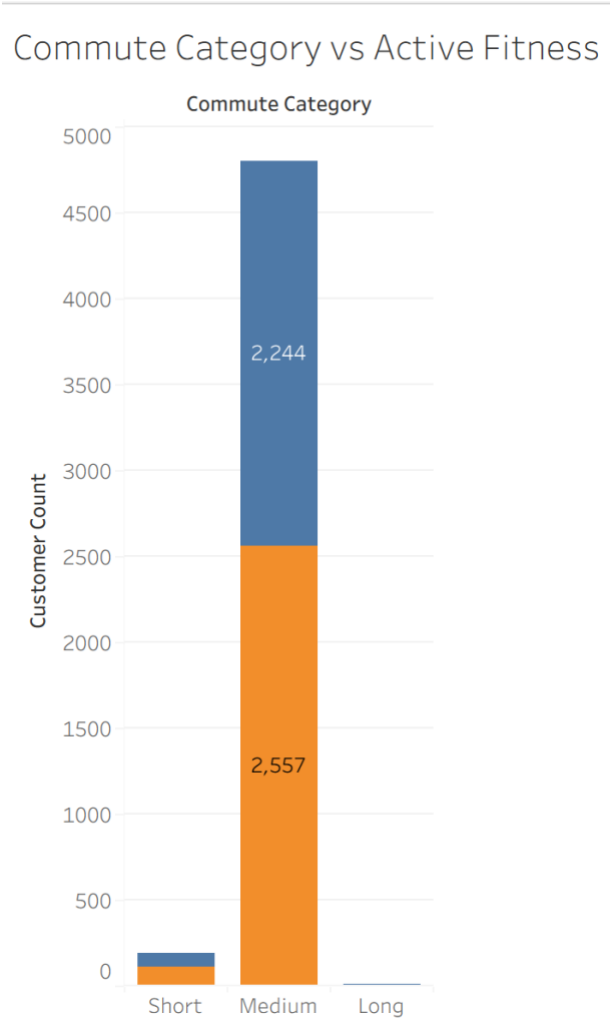


Figure 10: Commute Category vs Active Fitness

Overall, the visualizations reveal a valuable customer segment for TixScout defined by a strong tech engagement, a lot of streaming events, generally good economic circumstances, and in some cases, shared lifestyle attributes such as being pet owners or active. All of these pieces of information will help to develop more sophisticated, targeted, and personalized marketing campaigns that leverage the app to increase music event engagement and to generate ticket sales.

4 Insights and Recommendations

This project systematized and categorized TixScout's original customer dataset of 5,000 rows by 60 columns to bring a recommended direction to building marketing segmentation scenarios for use in the company's communications. By theoretically reducing features along with collapsing redundancies and the formation of ten new features fit to the context or scenario, the dataset now aligns with TixScout goals, is intelligible, and is less bulky. Columns that were offering low value to an increased understanding of the customer, such as a more in-depth breakdown of pet type, vehicle matters, and any political or voting frameworks, were reduced to eliminate 'noise'. Granular values were collapsed into larger segment indicators based on actionable and behavior-based judgments, such as the Pet Ownership Indicator or Commute Category Indicator, and new variables were engineered for scenario development, such as the Debt Stress Level, Financial Flexibility Score, and Tech Engagement Score, that guide audience segmentation and nuance.

The analysis was backed by ten polished Tableau visualizations, evenly divided between single variable and two variable charts, which provided key insights into customer behaviors and relationships. From those learning, I identified tech-savvy heavy streamers as ideal candidates to receive app-based marketing and VIP offers, financially comfortable high credit card users as prime candidates for premium ticket sales, and pet owners as a significant demographic to target for themed or pet-friendly promotions. Commute patterns also emerged as an important factor in determining attendance likelihood for weekday events.

The final dataset contains a total of 5,000 records of customer data, but was narrowed from 60 original columns to a simple, insight-oriented column. All the new features created from scenario-based data were colored yellow, and any changed or merged columns were highlighted green to identify and create traceability in the dataset. Once irrelevant details have been discarded, focus on high-impact and practical variables, and think of new features; this dataset is now ready for further analysis in the next stage of the project. It is a practical, business-useful tool usable by TixScout with data-driven customer segmentation strategies.